

CHAPTER 1

Tang
&
Rong

Rong
Tang

2019–
2025

An-
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of
Sta-
tis-
tics

MCMC

1.

1.1.

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 D GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

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Tang
&
Yang
2019–
2025

MCMC

1

$\mathcal{Z}$
VAE

$\mathcal{Z}$

MCMC

2.

2.1.

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$$\mathcal{M}^*$$

$$\mathbb{R}^D$$

$$d$$

$$\beta\text{-}$$

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0,\beta-1]$$

$$p^*$$

$$\mathbb{R}^{\mathcal{D}}$$

Lebesgue

Lebesgue

Par-
ti-
tion
of
Unity
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

$$?]]\text{tang2022minimax}$$

$$?]]\text{tang2023lowdim}$$

Reach

reach

$$\begin{aligned} \text{reach}(\mathcal{M}) = \\ \inf\{r > \\ 0 : \\ \mathcal{M} \\ r- \end{aligned}$$

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\mu - \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\nu \right|.$$

$\mathcal{F}$ γ -
Hölder

Sobolev

1-
LipschitzWasserstein-
1

RKHS

MMD
 γ

trade-
off

$$\gamma = 1$$

$$d_\gamma$$

Wasserstein-
1

$$\gamma \rightarrow 0^+$$

$$d_\gamma$$

?]tang2025regression

2.3.

•
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

β -

d

Haus-
dorff

D

Lep-
ski

chain-
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peel-
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cal-
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tion

Gir-
sanov

Poincaré

2.4.

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Gauss-
ian

Langevin

$$p_{\sigma}(x) = \int p_{\text{data}}(y) \phi_{\sigma}(x - y) \, dy$$

$$L_{\infty}$$

-

$$\nabla \log p_t(x)$$

$$t$$

$$t =$$

$$0$$

$$\frac{t}{T} =$$

?]tang2024diffusion

$$\mathcal{S}_{\text{NN}}$$

$$t$$

Girsanov

Gir-
sanov

SDE

3.

3.1. Tang
&
Yang
(2022)

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2022.

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GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

α

β

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unity

VAE

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Lep-
ski

Fano

Varshamov-
Gilbert

?]]tang2022minimax

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?]]tang2024diffusion

3.2. Tang
&
Yang
(2023a)

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ICML,
2023.

VAE
GAN

MLD-
MAE

K

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ρ_k

1-
Wasserstein

α -

β -

d

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big)\vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$$\begin{array}{l} \alpha\leq \\ \beta- \\ 1 \end{array}$$

/

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?]tang2022minimax

3.3. ?]]tang2023robust

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Rong

Tang,
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2023.

Stiefel

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Bayesian
RPE-
TEL

(1)

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(3)

Bernstein-
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(BvM)

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3.4. Tang
&
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(2023c)
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Rong
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2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
 ℓ_2

**Wasser-
stein**

1

2

$$O(n \log n + n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

**3.5. Tang
&
Yang
(2024a)**

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2024.

SOTA

Langevin

—

d D
Langevin**Gauss-
ian** $\tilde{O}(\sigma)$ σ n

d L_∞

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1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

?]tang2022minimax

?]tang2022minimax

3.6. `metropolis-adjusted-langevin`

Metropolis-
Adjusted
Langevin

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Rong
Tang
and
Yun
Yang.
“Metropolis-
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Langevin
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Be-
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ICML,
2024.

MALA

MCMC

$$O(d^{1/3})$$

s-
conductance
pro-
file

MALA

$O(d^{1/3})$

**burn-
in**

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file

**warm-
start**

$\log M_0$

$\log \log M_0$

**Tang
(2023b)**

**Tang
(2023b)**

BvM

BvM

MALA

3.7. Tang
&
Yang
(2025a)
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arXiv,
2025.

X

Y

$\mu_{Y|X}^*$

X Y

IPM

(1) **Regime**
1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2) **Regime**
2

:

$$\mathcal{M}_{Y|X}$$

$$X$$

$$\begin{array}{l} (3) \quad n^{-\gamma\beta_Y/d_Y} \\ \text{Regime} \\ 3 \end{array}$$

$$:$$

$$\mathcal{M}_{Y|x}$$

$$x$$

$$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$$

γ -
Hölder
IPM
 d_γ

$$\gamma = 0$$

1-
Wasserstein

$$\gamma = 1$$

3.8. Tang
&
Yang
(2025b)
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Rong
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Yun
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and
Manifold-
Adaptive,”
arXiv,
2025.

?]]tang2024diffusion

(1)

$$\alpha_X \in [0, 1]$$

(2)

X

Y

Wasser-
stein

$$(d_X, d_Y)$$

$$(D_X, D_Y)$$

lo-
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peel-
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tech-
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Gir-
sanov

?]]tang2024diffusion,
?]]tang2025regression

?]]tang2024diffusion

→
?]]tang2025regression

→

4.

Tang-
Yang

2019—
2025